Agile Theory Course Project

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Course title: Agile Development Process and Devops

Slot: C1+TC1

Course code: ISWE406P

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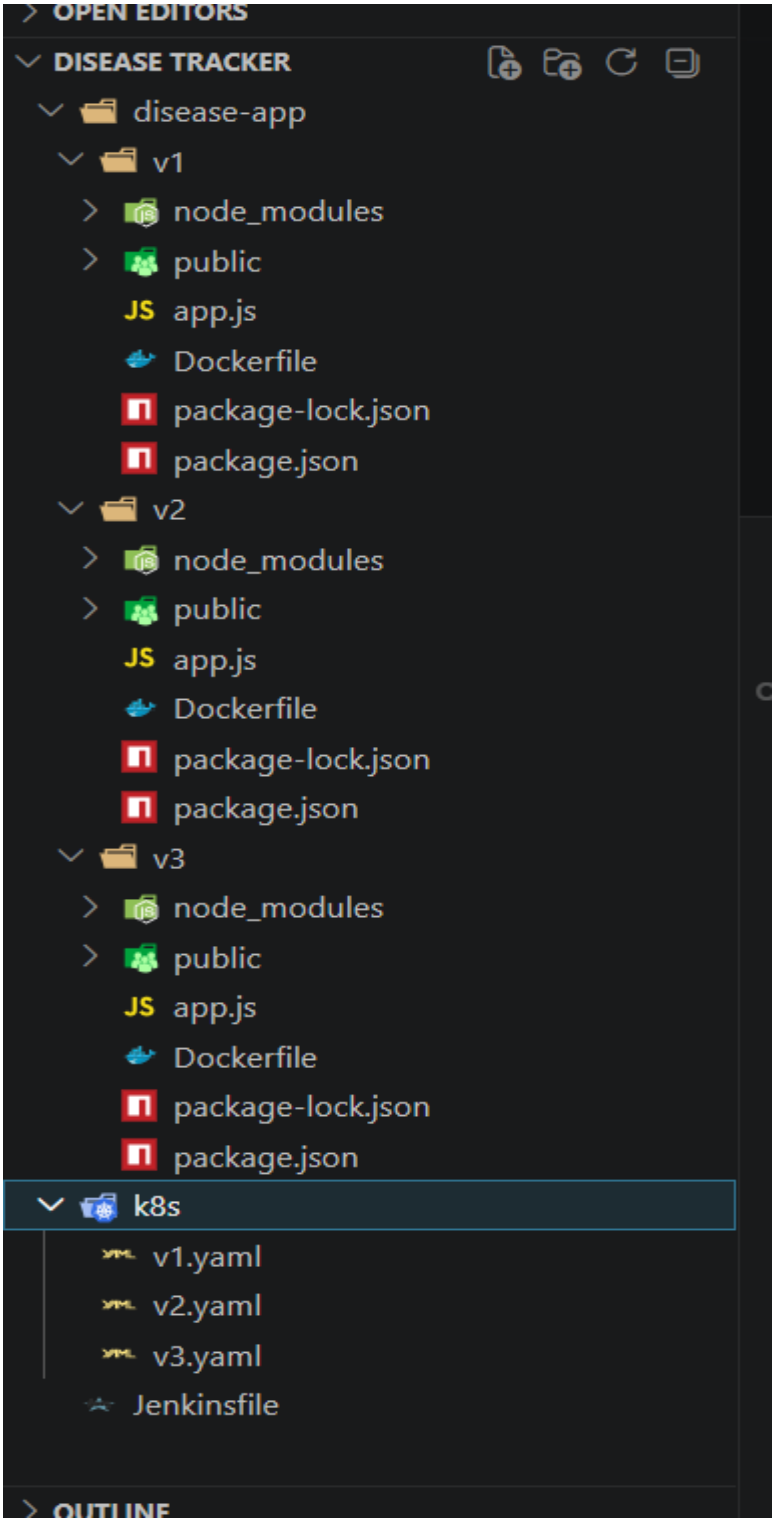
Topic: Disease Tracker

Disease Tracker System

Disease Tracker System with Add (Enter case data), View (Display cases), and Filter (Sort by location/date) features developed in three versions (v1, v2, v3) using VS Code, managed via GitHub, built with Jenkins pipeline, containerized using Docker (tagged v1, v2, v3), pushed to DockerHub, and deployed in Kubernetes as 3 separate deployments/services with each version accessible on different localhost ports.

The application developed in three versions (v1, v2, v3).



v1: **with Add (Enter case data)**

Public: index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-
width,initial-scale=1" />
    <title>Disease Tracker v1 - Add Cases</title>
    <style>
      body {
        font-family: Arial, Helvetica, sans-serif;
        padding: 20px;
      }
      label {
        display: block;
        margin-top: 8px;
      }
    </style>
  </head>
  <body>
    <h1>Disease Tracker v1 — Add Case</h1>
    <form id="caseForm">
      <label>Disease name: <input type="text" id="disease"
required /></label>
      <label>Location: <input type="text" id="location" required
/></label>
      <button type="submit">Add Case</button>
    </form>
    <p id="status"></p>

    <script src="script.js"></script>
  </body>
</html>
```

Script.js

```
// v1: simple form sends POST /add to server to add a case
```

```
document.getElementById('caseForm').addEventListener('submit', async (e) => {
  e.preventDefault();
  const disease = document.getElementById('disease').value.trim();
  const location = document.getElementById('location').value.trim();
  const status = document.getElementById('status');
  try {
    const res = await fetch('/add', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify({ disease, location })
    });
    const json = await res.json();
    if (res.ok) {
      status.textContent = `Added. Total cases on server: ${json.total}`;
    } else {
      status.textContent = `Error: ${json.error || 'unknown'}`;
    }
  } catch (err) {
    status.textContent = 'Request failed: ' + err.message;
  }
});
```

App.js

```
// Simple Express server for Disease Tracker v1
// Features: serve static front-end and accept POST /add to add a case (in-memory)
const express = require('express');
const path = require('path');
const app = express();
const PORT = process.env.PORT || 3000;

app.use(express.json());
app.use(express.urlencoded({ extended: true }));

// Serve static files from ./public
app.use(express.static(path.join(__dirname, 'public')));
```

```
// In-memory store for v1 (simple lab-level storage)
const cases = [];

// Add a new case (disease + location)
app.post('/add', (req, res) => {
  const { disease, location } = req.body;
  if (!disease || !location) {
    return res.status(400).json({ error: 'Missing disease or location' });
  }
  cases.push({ disease, location, addedAt: new Date().toISOString() });
  return res.json({ success: true, total: cases.length });
});

// Return all cases (for debugging/testing)
app.get('/cases', (req, res) => res.json(cases));

app.listen(PORT, () => console.log(`Disease Tracker v1 running on port ${PORT}`));
```

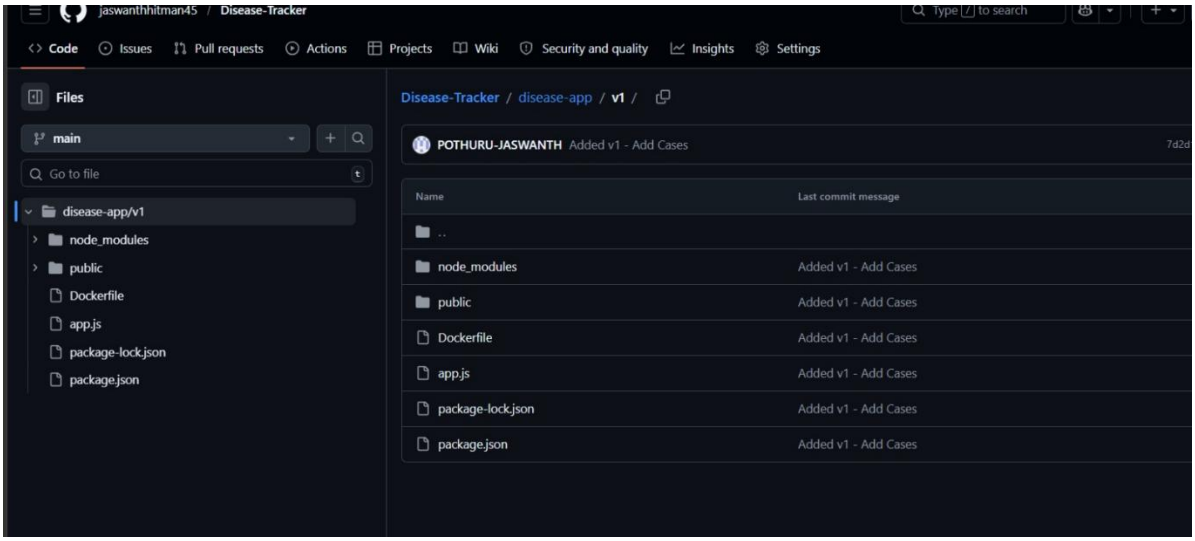
Dockerfile

```
FROM node:18
WORKDIR /app
COPY package*.json ./
RUN npm install COPY . .
EXPOSE 3000
CMD ["npm", "start"]
```

Pushing v1 into GitHub:

```
  create mode 100644 disease-app/v1/public/script.js
  PS C:\Users\Lenovo\Desktop\Disease Tracker> git push -u origin main
  Enumerating objects: 721, done.
  Counting objects: 100% (721/721), done.
  Delta compression using up to 8 threads
  Compressing objects: 100% (666/666), done.
  Writing objects: 100% (721/721), 719.68 KiB | 2.07 MiB/s, done.
  Total 721 (delta 126), reused 0 (delta 0), pack-reused 0 (from 0)
  remote: Resolving deltas: 100% (126/126), done.
  To https://github.com/jaswanthhitman45/Disease-Tracker.git
   * [new branch]      main -> main
  branch 'main' set up to track 'origin/main'.
```

Pushed into Github:



V2:

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-
width,initial-scale=1" />
    <title>Disease Tracker v2 - View Cases</title>
    <style>
      body {
        font-family: Arial;
        padding: 20px;
      }
      label {
        display: block;
        margin-top: 8px;
      }
      table {
        border-collapse: collapse;
        margin-top: 12px;
```

```
      }
      td,
      th {
        border: 1px solid #ccc;
        padding: 6px;
      }
    </style>
  </head>
  <body>
    <h1>Disease Tracker v2 — Cases (localStorage)</h1>
    <form id="caseForm">
      <label>Disease name: <input type="text" id="disease"
required /></label>
      <label>Location: <input type="text" id="location" required
/></label>
      <button type="submit">Add / Increment</button>
    </form>
    <p id="status"></p>

    <h2>Stored Cases</h2>
    <div id="casesContainer"></div>

    <script src="script.js"></script>
  </body>
</html>
```

script.js

```
// v2: store cases in localStorage and update quantity if same disease+location
const STORAGE_KEY = 'disease_cases_v2';

function loadCases() {
  const raw = localStorage.getItem(STORAGE_KEY);
```

```
  return raw ? JSON.parse(raw) : [];
}

function saveCases(list) {
  localStorage.setItem(STORAGE_KEY, JSON.stringify(list));
}

function render() {
  const list = loadCases();
  const container = document.getElementById('casesContainer');
  if (list.length === 0) {
    container.innerHTML = '<i>No cases stored yet.</i>';
    return;
  }
  let html = '<table><tr><th>Disease</th><th>Location</th><th>Quantity</th></tr>';
  list.forEach(c => {
    html += `<tr><td>${escapeHtml(c.disease)}</td><td>${escapeHtml(c.location)}</td><td>${c.quantity}</td></tr>`;
  });
  html += '</table>';
  container.innerHTML = html;
}

function escapeHtml(s){ return String(s).replace(/&/g,'&amp;').replace(/</g,'&lt;').replace(/>/g,'&gt;'); }

document.getElementById('caseForm').addEventListener('submit', (e)=>{
  e.preventDefault();
  const disease = document.getElementById('disease').value.trim();
  const location = document.getElementById('location').value.trim();
  const status = document.getElementById('status');
  if(!disease||!location){ status.textContent='Enter disease and location'; return; }

  const list = loadCases();
  const idx = list.findIndex(x=>x.disease===disease && x.location===location);
  if(idx>=0){
    list[idx].quantity += 1;
    status.textContent = `Incremented quantity to ${list[idx].quantity}`;
```

```
  } else {
    list.push({ disease, location, quantity: 1 });
    status.textContent = 'Added new case (quantity 1)';
  }
  saveCases(list);
  render();
});

// initial render
render();
```

app.js

```
// Simple Express server for Disease Tracker v2
// Features: serve static front-end. Front-end stores cases in localStorage and updates quantities.
const express = require('express');
const path = require('path');
const app = express();
const PORT = process.env.PORT || 3000;

app.use(express.json());
app.use(express.urlencoded({ extended: true }));
app.use(express.static(path.join(__dirname, 'public')));

// Simple health route
app.get('/health', (req, res) => res.send({ status: 'ok' }));

app.listen(PORT, () => console.log(`Disease Tracker v2 running on port ${PORT}`));
```

Dockerfile

```
FROM node:18
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
```

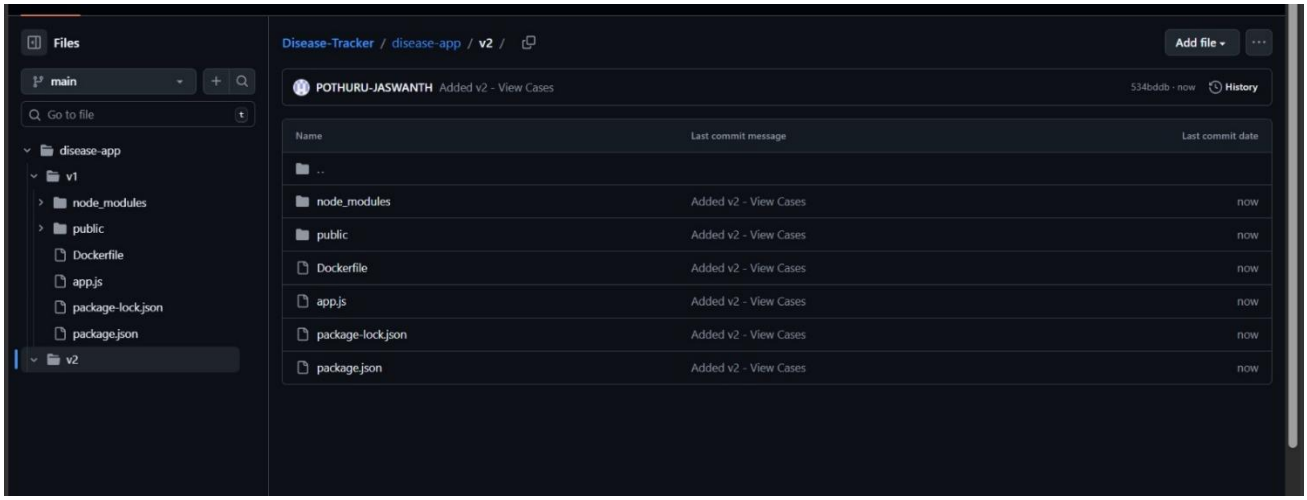
EXPOSE 3000

CMD ["npm", "start"]

Pushing v2 into GitHub:

```
 create mode 100644 disease-app/v2/public/script.js
PS C:\Users\Lenovo\Desktop\Disease Tracker> git push
Enumerating objects: 723, done.
Counting objects: 100% (723/723), done.
Delta compression using up to 8 threads
Compressing objects: 100% (667/667), done.
Writing objects: 100% (721/721), 719.93 KiB | 1.98 MiB/s, done.
Total 721 (delta 126), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (126/126), done.
To https://github.com/jaswanthhitman45/Disease-Tracker.git
   7d2d18b..534bddb  main -> main
PS C:\Users\Lenovo\Desktop\Disease Tracker>
```

Pushed into Github:



V3:

index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width,initial-
scale=1" />
    <title>Disease Tracker v3 - Filter Cases</title>
    <style>
```

```
      body {
        font-family: Arial;
        padding: 20px;
      }
      label {
        display: block;
        margin-top: 8px;
      }
      table {
        border-collapse: collapse;
        margin-top: 12px;
      }
      td,
      th {
        border: 1px solid #ccc;
        padding: 6px;
      }
    </style>
  </head>
  <body>
    <h1>Disease Tracker v3 — Filter by Location</h1>
    <form id="caseForm">
      <label>Disease name: <input type="text" id="disease"
required /></label>
      <label>Location: <input type="text" id="location" required
/></label>
      <button type="submit">Add / Increment</button>
    </form>
    <p id="status"></p>

    <h2>Filter</h2>
    <label
      >Location filter:
      <input
        type="text"
        id="filterLocation"
        placeholder="Enter location to filter"
    /></label>
    <button id="clearFilter">Clear Filter</button>
```

```
    <h2>Stored Cases</h2>
    <div id="casesContainer"></div>

    <script src="script.js"></script>
  </body>
</html>
```

app.js

```
 // Simple Express server for Disease Tracker v3
// Features: serve static front-end. Front-end
stores cases in localStorage and supports filtering
by location.
const express = require('express');
const path = require('path');
const app = express();
const PORT = process.env.PORT || 3000;

app.use(express.json());
app.use(express.urlencoded({ extended: true }));
app.use(express.static(path.join(__dirname,
'public')));

app.get('/health', (req, res) => res.send({ status:
'ok' }));

app.listen(PORT, () => console.log(`Disease
Tracker v3 running on port ${PORT}`));
```

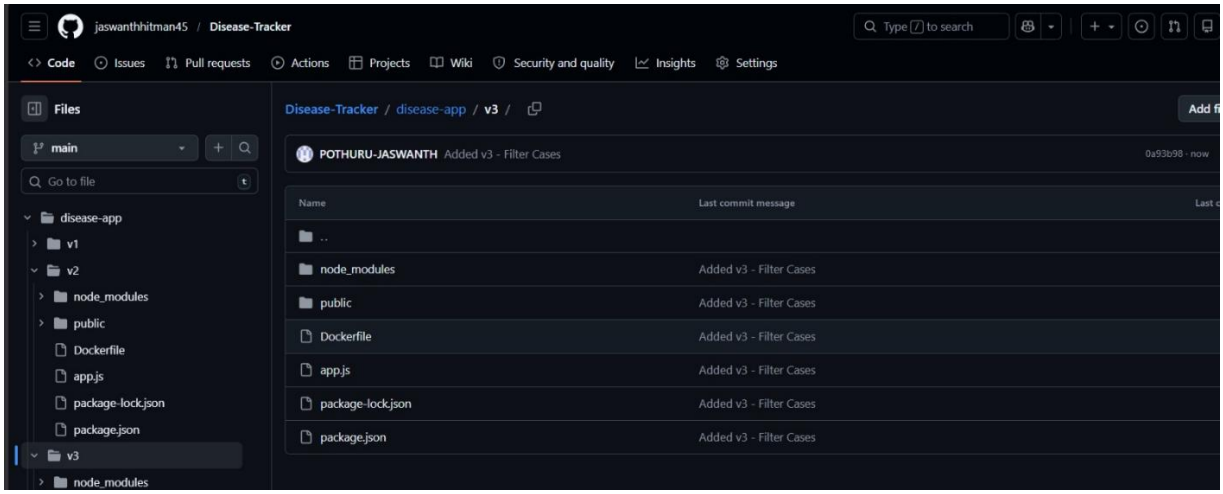
Dockerfile

```
FROM node:18
WORKDIR /app
COPY package*.json ./
RUN npm install COPY . .
EXPOSE 3000 CMD ["npm", "start"]
```

Pushing v3 into GitHub:

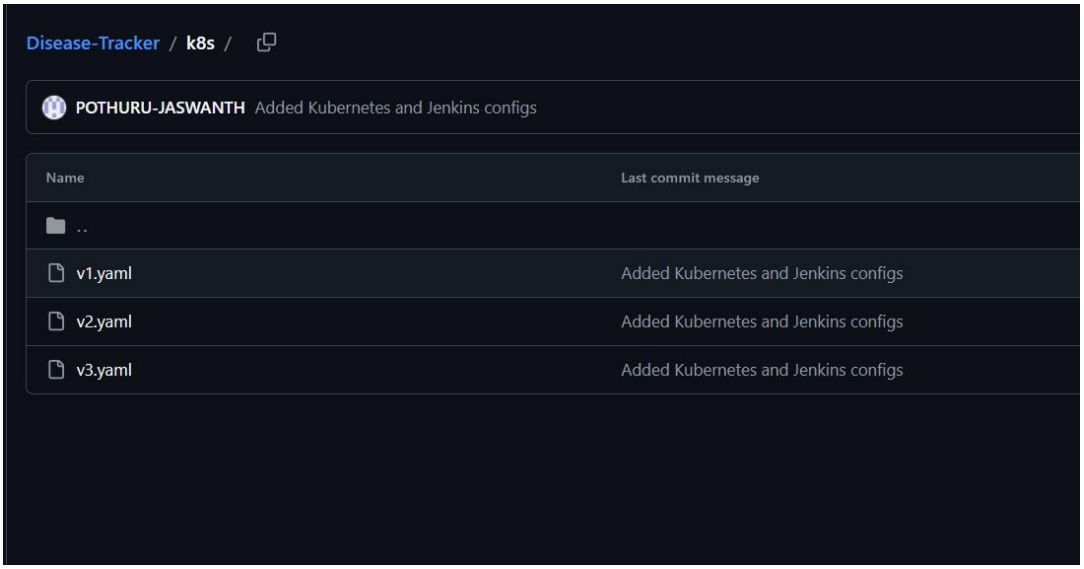
```
  create mode 100644 disease-app/v3/public/index.html
  create mode 100644 disease-app/v3/public/script.js
PS C:\Users\Lenovo\Desktop\Disease Tracker> git push
Enumerating objects: 723, done.
Counting objects: 100% (723/723), done.
Delta compression using up to 8 threads
Compressing objects: 100% (667/667), done.
Writing objects: 100% (721/721), 720.06 KiB | 1.68 MiB/s, done.
Total 721 (delta 126), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (126/126), done.
To https://github.com/jaswanthhitman45/Disease-Tracker.git
   534bddb..0a93b98  main -> main
PS C:\Users\Lenovo\Desktop\Disease Tracker>
Ln 14, Col 1
```

Pushed into Github:



k8s Pushed into Github:

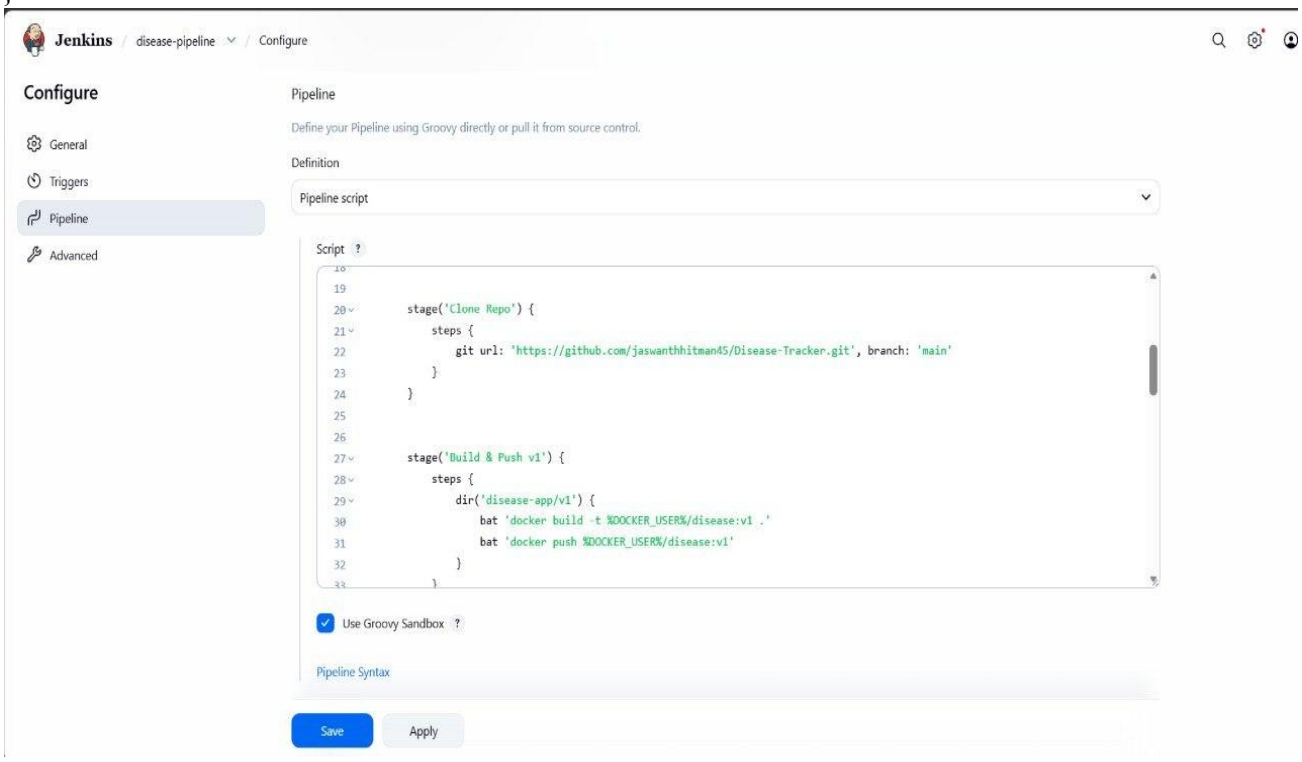
```
PS C:\Users\Lenovo\Desktop\Disease Tracker> git add k8s Jenkinsfile
PS C:\Users\Lenovo\Desktop\Disease Tracker> git commit -m "Added Kubernetes and Jenkins c
onfigs"
[main ee2ae1c] Added Kubernetes and Jenkins configs
 4 files changed, 168 insertions(+)
 create mode 100644 Jenkinsfile
 create mode 100644 k8s/v1.yaml
 create mode 100644 k8s/v2.yaml
 create mode 100644 k8s/v3.yaml
PS C:\Users\Lenovo\Desktop\Disease Tracker> git push
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1.23 KiB | 418.00 KiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/jaswanthhitman45/Disease-Tracker.git
   0a93b98..ee2ae1c  main -> main
PS C:\Users\Lenovo\Desktop\Disease Tracker>
Ln 14, Col 1   Spaces: 4   UTF-8   CR
```

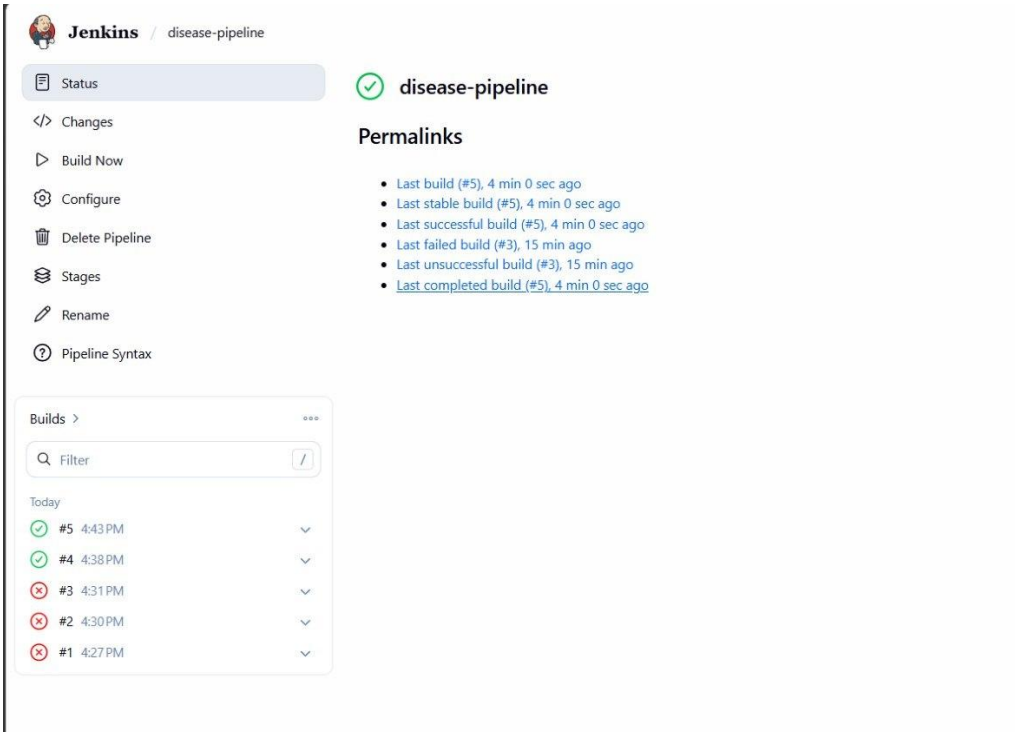


Pipeline Script:

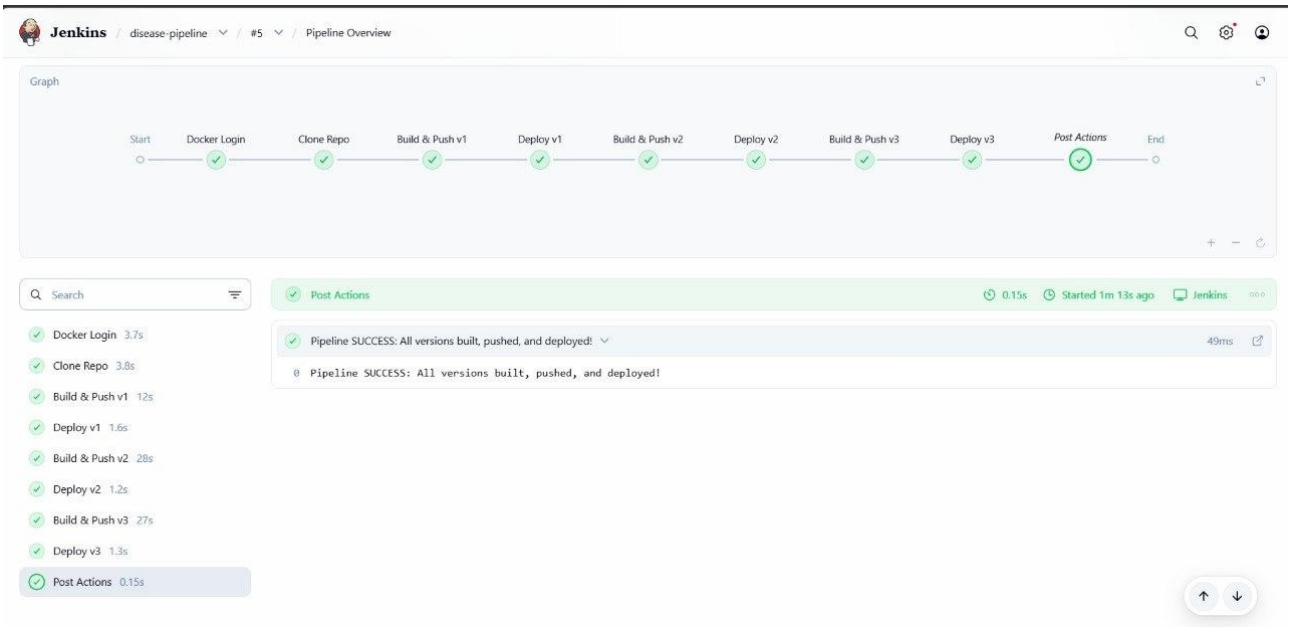
```
pipeline {
   agent any

   environment {
      DOCKER_USER = "pothurujaswanth"
      KUBECONFIG = "C:\\Users\\vigne\\.kube\\config"
   }

   stages {

      stage('Clone Repo') {
         steps {
            git url: 'https://github.com/jaswanthhitman45/Disease-Tracker.git', branch: 'main'
         }
      }

      stage('Build & Push v1') {
         steps {
            dir('disease-app/v1') {
               bat 'docker build -t %DOCKER_USER%/disease:v1 .'
               bat 'docker push %DOCKER_USER%/disease:v1'
            }
         }
      }

      stage('Deploy v1') {
         steps {
            bat 'kubectl apply -f k8s\\v1.yaml --validate=false'
         }
      }

      stage('Build & Push v2') {
         steps {
            dir('disease-app/v2') {
               bat 'docker build -t %DOCKER_USER%/disease:v2 .'
               bat 'docker push %DOCKER_USER%/disease:v2'
            }
         }
      }
```

```
        stage('Deploy v2') {
            steps {
                bat 'kubectl apply -f k8s\\v2.yaml --validate=false'
            }
        }

        stage('Build & Push v3') {
            steps {
                dir('disease-app/v3') {
                    bat 'docker build -t %DOCKER_USER%/disease:v3 .'
                    bat 'docker push %DOCKER_USER%/disease:v3'
                }
            }
        }

        stage('Deploy v3') {
            steps {
                bat 'kubectl apply -f k8s\\v3.yaml --validate=false'
            }
        }
    }
}
```





Jenkins Pipeline Stages View:



Jenkins / disease-pipeline / #5

```
14f3b71afb7e: Pushed
v3: digest: sha256:4ef522b4c64c2280bc8937d0292b8657802ae97170628f5c34018696a92bbd0a size: 856
[Pipeline] }
[Pipeline] // dir
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy v3)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\disease-pipeline>kubectl apply -f k8s\v3.yaml --validate=false
deployment.apps/disease-v3 created
service/disease-v3-svc created
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline SUCCESS: All versions built, pushed, and deployed!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Pushed to DockerHub:

hub | Explore | My Hub | Sear

pothurujaswanth
Docker Personal

- Repositories
- Hardened Images
- Collaborations
- Settings
  - Default privacy
  - Notifications
- Billing
- Usage
  - Pulls
  - Storage

Repositories / disease / General

pothurujaswanth/disease

Last pushed about 1 hour ago · Repository size: 49 MB · 0 · 33

Add a description

Add a category

General | Tags | Image Management | Collaborators | Webhooks | Settings

Tags

DOCKER SCOUT INACTIVE
Activate

This repository contains 3 tag(s).

Tag	OS	Type	Pulled	Pushed
v3		Image	less than 1 day	about 1 hour
v2		Image	less than 1 day	about 1 hour
v1		Image	less than 1 day	about 1 hour

See all

Docker Hub showing cart tags v1, v2, and v3:

v1:

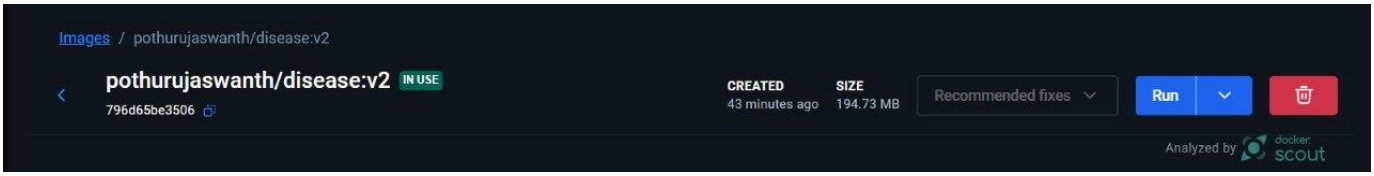
Images / pothurujaswanth/disease:v1

pothurujaswanth/disease:v1 IN USE
0763b09d1002

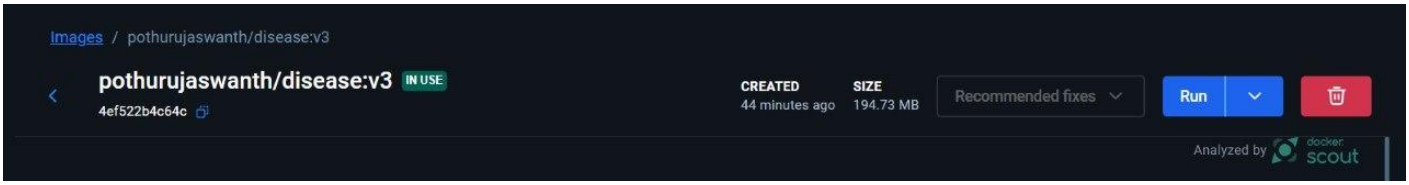
CREATED 60 minutes ago | SIZE 194.73 MB | Recommended fixes | Run

Analyzed by docker scout

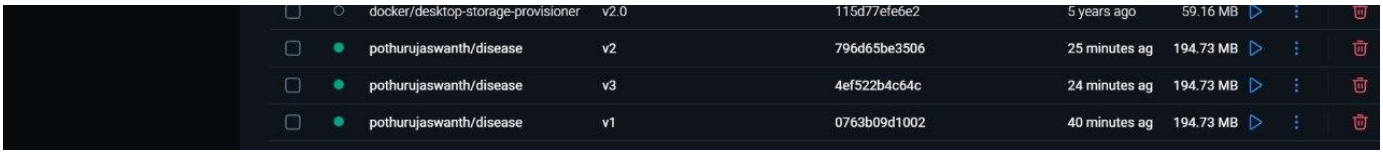
v2:



v3:



Images in docker.destop:



docker/desktop-storage-provisioner	v2.0	115d77efe6e2	5 years ago	59.16 MB	
pothurujaswanth/disease	v2	796d65be3506	25 minutes ag	194.73 MB	
pothurujaswanth/disease	v3	4ef522b4c64c	24 minutes ag	194.73 MB	
pothurujaswanth/disease	v1	0763b09d1002	40 minutes ag	194.73 MB	

Deployed in Kubernetes as 3 separate deployments/services:

```
PS C:\Users\vigne\Desktop\Disease-Tracker> kubectl get pods
NAME                                    READY   STATUS    RESTARTS      AGE
conference-deployment-67cb6dbb4d-gcr42  1/1     Running   1 (56m ago)   3d2h
conference-deployment-67cb6dbb4d-gtmkg  1/1     Running   1 (56m ago)   3d2h
disease-v1-5956f9c6dd-lxvl9             1/1     Running   0             5m50s
disease-v2-b89f86685-66hvj              1/1     Running   0             5m21s
disease-v3-777c7f987d-558tk             1/1     Running   0             5m15s
```

Version – 1

```
PS C:\Users\vigne\Desktop\Disease-Tracker> kubectl port-forward service/disease-v1-svc  8881:3000
Forwarding from 127.0.0.1:8881 -> 3000
Forwarding from [::1]:8881 -> 3000
Handling connection for 8881
Handling connection for 8881
Handling connection for 8881
```

Version-2:

```
PS C:\Users\vigne\Desktop\Disease-Tracker> kubectl port-forward service/disease-v2-svc  8881:3000
Forwarding from 127.0.0.1:8881 -> 3000
Forwarding from [::1]:8881 -> 3000
Handling connection for 8881
Handling connection for 8881
```

localhost:8881

Disease Tracker v2 — Cases (localStorage)

Disease name: cough

Location: vellore

Add / Increment

Added new case (quantity 1)

Stored Cases

Disease	Location	Quantity
asthama	vellore	1
flu	vellore	1
cough	vellore	1

Version - 3

```
PS C:\Users\vigne\Desktop\Disease-Tracker> kubectl port-forward service/disease-v3-svc  8881:3000
Forwarding from 127.0.0.1:8881 -> 3000
Forwarding from [::1]:8881 -> 3000
Handling connection for 8881
Handling connection for 8881
```

localhost:8881

Disease Tracker v3 — Filter by Location

Disease name:

Location:

Add / Increment

Filter

Location filter: Enter location to filter

Clear Filter

Stored Cases

No cases match.

Disease Tracker v3 — Filter by Location

Disease name: asthama

Location: ongole

Add / Increment

Added new case (quantity 1)

Filter

Location filter: vellore

Clear Filter

Stored Cases

Disease	Location	Quantity
asthama	vellore	1

localhost:8881

Disease Tracker v3 — Filter by Location

Disease name: asthama

Location: ongole

Add / Increment

Added new case (quantity 1)

Filter

Location filter: Enter location to filter

Clear Filter

Stored Cases

Disease	Location	Quantity
asthama	vellore	1
asthama	kurnool	1
asthama	ongole	1